

GitUbe Graph (gitube)

The Italian Olympic staff is very active on *GitUbe*, a website that is used by a lot of developers around the world to collaborate on open source code. William likes GitUbe so much that he wrote a “GitUbe-like” system, for his personal use. Sadly, a key feature is missing in William’s system: the *GitUbe Graph*.

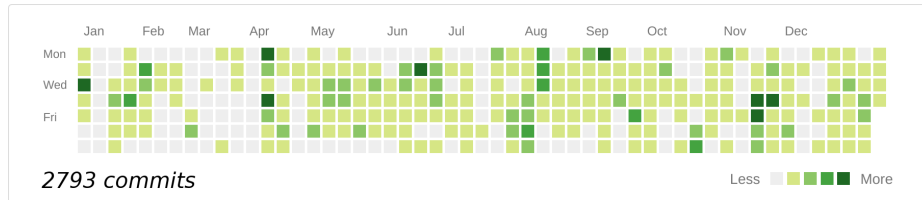


Figure 1: A sample *GitUbe Graph*.

The “GitUbe Graph” is a neat way to show users’ activity. For each user, the graph shows a grid where each cell is colored with some color depending on the number of *commits* made by the user in that day (e.g. dark green for a very active day, light green for a less active day, and so on). The grid has 7 rows (one for each day of the week) and some amount of columns (depending on how many days we want to show; GitUbe, for example, always shows the whole year).

For the purpose of this task, we can assume:

- We want the graph to show D days (numbered from 1 to D).
- The first day (or “day number 1 of D ”) is always a Monday.
- The week starts on Monday.
- Like GitUbe, we will number the days in the graph going column by column, starting from the top left cell of the grid and ending in the bottom right cell.
- To better support colorblind people, we won’t use colors at all, and instead use special symbols:
 - The ‘.’ symbol, when there are less than 10 commits in the day.
 - The ‘o’ letter, when there are from 10 to 19 commits in the day.
 - The ‘0’ letter, when there are 20 or more commits in the day.

This is an example graph:

```
0o00oo.o000.00.00o..0..0.00.0.ooo0.ooo..
o00000000000.o..o0o0o0o0.0..0o.oo.ooo.o0
.o..o0.o.0o.0o..o.000.0.0.0.o0.o.o.o.o0
..00o00.0.0o0o0o..00..o..o0ooo00.o0o0.
.0o0o0o0ooo00..ooo0ooo.o0...o.00.0o0.o0
000ooo00ooo0.o.0.o.o000oo...00..000o..0.
00.00ooo00...0.oo.o.00..o...0o..0.o.o00
```

You will be given the number of days D , the number of commits N made by some user, and then the actual list of commits. For each commit, you are given a single integer C_i which represents the day in which the commit was made.

Help William compute the user graph!

📎 Among the attachments of this task you may find a template file `github.*` with a sample incomplete implementation.

Input

The first line contains two integers D and N . The second line contains N integers C_i .

Output

You need to print the user graph as previously described.

Constraints

- $1 \leq D \leq 366$ (it might be a leap year, or a 1-day long period, and so on).
- $1 \leq N \leq 100\,000$.
- $1 \leq C_i \leq D$ for each $i = 0 \dots N - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** [5 points]: Examples.
- **Subtask 2** [25 points]: $N \leq 1000$.
- **Subtask 3** [40 points]: $D \leq 7$.
- **Subtask 4** [30 points]: No additional limitations.

Examples

input.txt	output.txt
<pre> 20 37 4 8 15 5 5 13 13 5 3 11 13 13 16 18 5 13 13 1 13 18 13 13 5 5 12 4 5 8 5 6 4 14 9 13 5 1 5 </pre>	<pre> o.. .o. .. </pre>
<pre> 7 40 1 2 6 2 2 4 2 2 1 2 6 2 6 1 4 6 2 7 2 2 2 2 7 3 2 3 1 2 2 4 2 2 2 4 2 2 6 4 5 2 </pre>	<pre> . 0 </pre>